



**S. B. JAIN INSTITUTE OF TECHNOLOGY,
MANAGEMENT & RESEARCH, NAGPUR.**

Session-2025-26 (Even)

TAE-I Project Based Learning

AI-Based Intrusion Detection System using Probabilistic Learning

Submitted By

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AI Technique(s) (2)	Problem Representation (4)	Tools Explored (2)	Outcome (2)	Total (10)
Sign of Student:			Sign of course In-charge :	

1. Industry Problem Context

In the IT industry, the problem of finding an **optimal route between locations** appears in:

- Navigation and map services (Google Maps–like systems)
- Logistics and delivery platforms
- Cloud network routing
- Data center traffic optimization
- Ride-sharing applications

The core objective is to **minimize cost**, such as **distance, time, bandwidth, or latency**.

2. Feasibility in IT Systems

- Real-world locations, servers, or nodes are modeled as **graphs**.
- Nodes represent **locations/routers/servers**.
- Edges represent **roads, links, or network connections**.
- Edge weights represent **distance, time, congestion, or cost**.
- Large-scale graphs are stored using **graph databases** or in-memory structures.

Hence, graph-based simulation is **highly feasible and scalable** in industry.

3. AI Techniques Used in Industry

- **A* Search**
 - Used for **fast and optimal path finding**
 - Combines actual cost + heuristic estimate
- **Best First Search**
 - Used when **speed is prioritized over optimality**
- **Heuristics Used**
 - Straight-line distance (maps)
 - Estimated latency (networks)
 - Historical traffic patterns (advanced systems)

These techniques ensure **real-time performance**.

4. Representation/Modeling in Industry Systems

- **State Space Graph**
 - Each state = current location/node
 - Transitions = possible next locations
- Graphs are:
 - Dynamically updated (traffic, failures)
 - Distributed across systems
- Often implemented using **adjacency lists** or **graph engines**.

5. Tools & Technologies in Industry

- **Python / Java / C++** for algorithm implementation
- **Graph frameworks** (e.g., internal graph engines, NetworkX for prototyping)
- **Cloud services** for large-scale computation
- **APIs** to fetch real-time data (traffic, network load)

Python + NetworkX is commonly used for **prototyping and proof-of-concept**, before deployment in production systems.

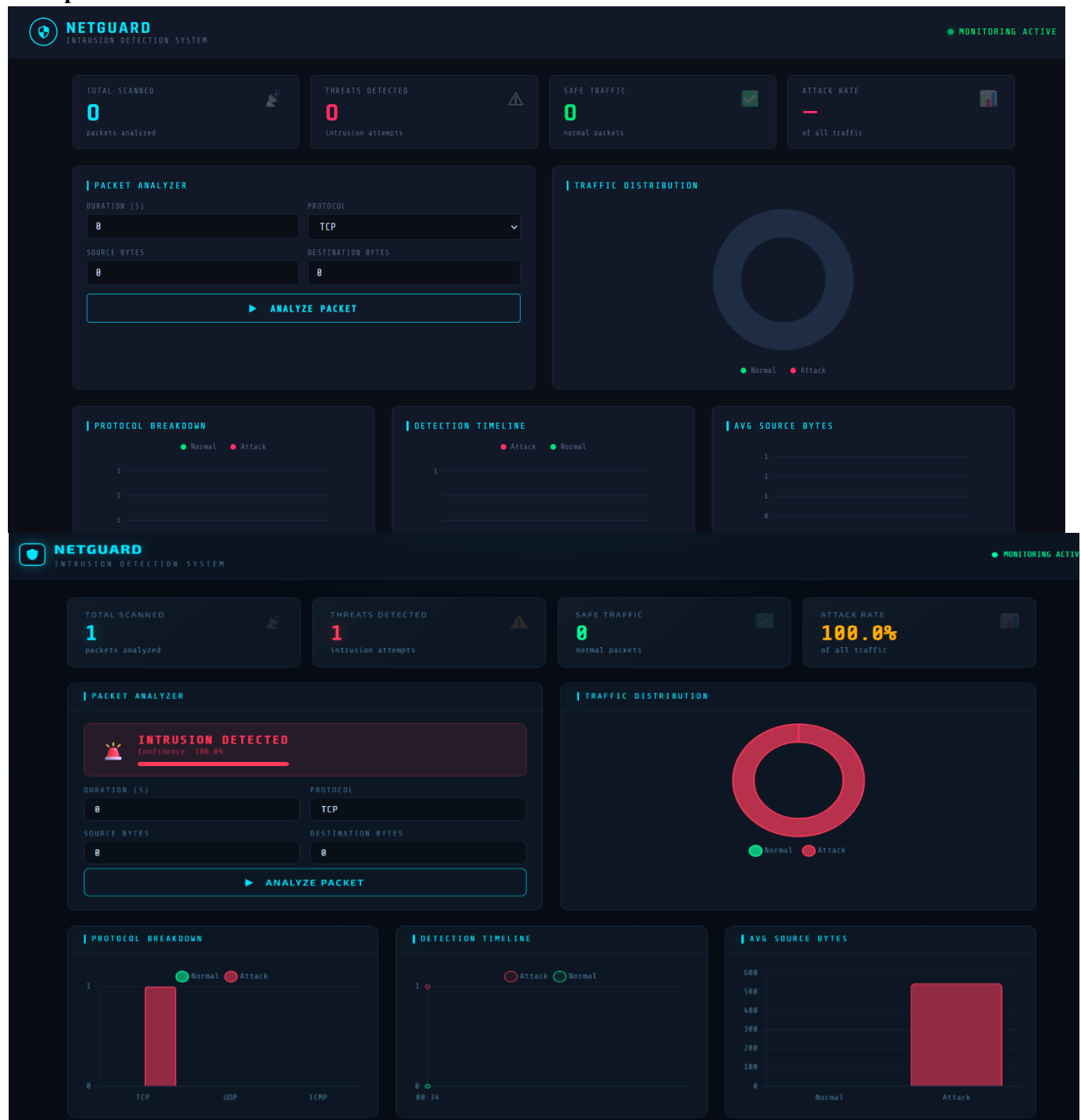
6. Industry-Level Outcome

- Optimal route computation in milliseconds
- Comparison of multiple routing strategies
- Reduced cost and improved efficiency
- Performance metrics such as:
 - Path cost
 - Execution time
 - Number of nodes expanded

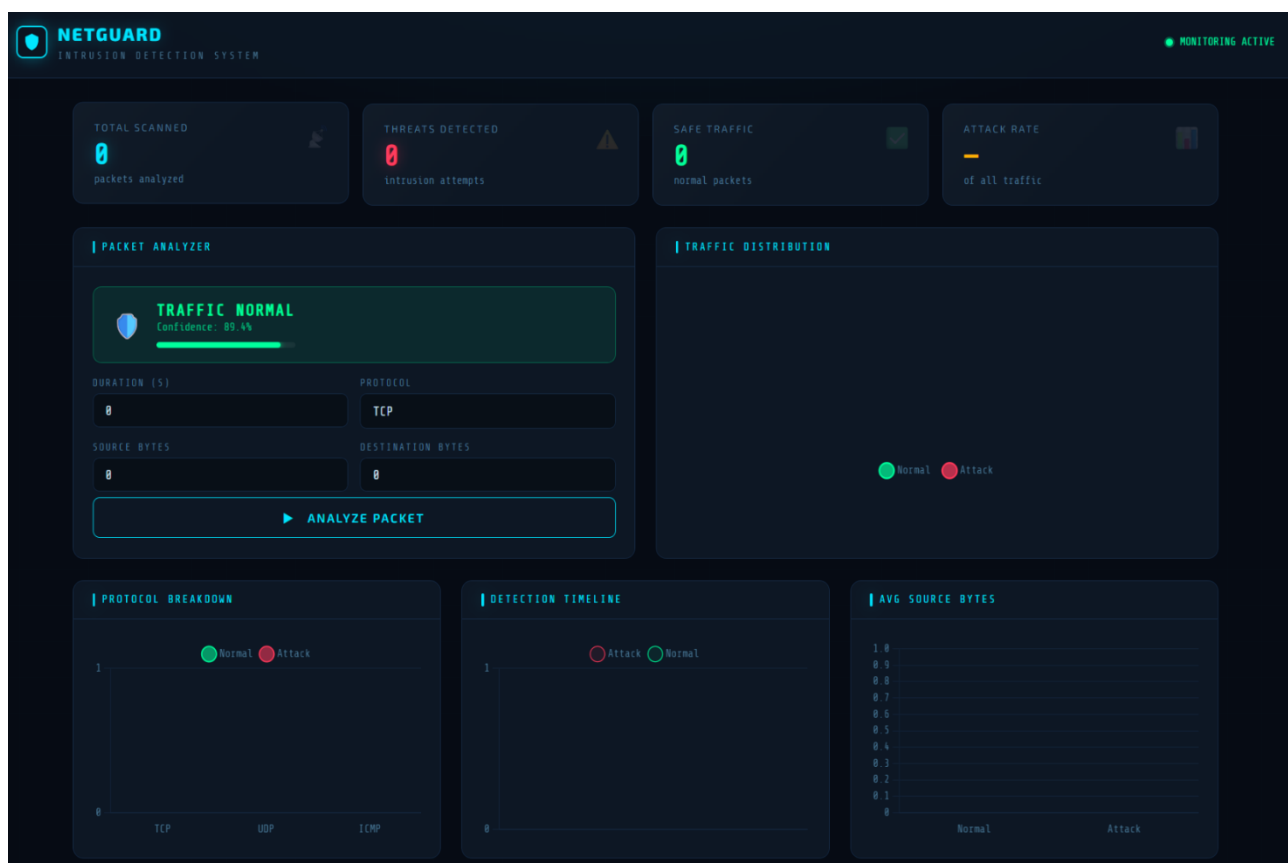
7. Real-World Industry Example/use

- **Navigation Apps:** Find fastest route avoiding traffic
- **Logistics Companies:** Optimize delivery routes
- **Cloud Providers:** Route data packets efficiently
- **IT Service Providers:** Optimize workflow or data pipeline paths

8. Implementation & Result



Screenshot 1: System Interface – Input Network Parameters



Screenshot 2: Prediction Result – Normal / Attack Classification

9. References

1. T. Mitchell, Machine Learning, McGraw-Hill, 1997.
2. KDD Cup 1999 Dataset – UCI Machine Learning Repository.
3. Scikit-learn Documentation – <https://scikit-learn.org>
4. S. J. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, Prentice Hall, 2020.